

Aditya Kumar Singh

Android Developer

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SUMMARY

Android and Flutter developer focused on building real-time, scalable mobile applications using Firebase, maps, and client-server systems. Experienced in developing production-style Android apps, optimizing app performance, and implementing real-time features for practical use cases.

TECHNICAL SKILLS

Languages: Java, Dart, Python, C

Mobile: Android (Java, XML), Flutter

Backend/DB: Firebase, MySQL, SQLite, Supabase

Core: DSA, OOP, REST APIs, Client-Server, Real-time systems

ML/NLP: LSTM, Attention, NLP pipelines (PyTorch)

Tools: Git, GitHub, Android Studio, IntelliJ, VS Code

PROFESSIONAL EXPERIENCE

Research Intern — Jadavpur University (Kolkata, West Bengal) | [Github](#) | [Documentation](#)

May 2025 – Jul 2025

- Architected EVX, a real-time EV-to-EV charging prototype enabling dynamic energy sharing between nearby vehicles
- Designed and implemented geospatial matching using OpenStreetMap (OSM) to connect donor/acceptor EVs within a ~5 km radius
- Built a low-latency backend using Firebase for real-time request handling and live tracking (sub-second updates)
- Conducted EV ecosystem analysis and identified infrastructure gaps
- Documented system architecture, workflows, and research findings

App Development Intern — CodeAlpha (Virtual) | [Github](#) |

Feb 2026

- Developed multiple Android applications (fitness tracker, quiz app, quote generator)
- Implemented local data persistence using Room Database and SharedPreferences
- Debugged and optimized app performance across modules

PROJECTS

Auctora — Real-Time Bidding Platform | [Github](#) |

- Built real-time auction rooms with Firebase (~<1s update latency) and Implemented full flow: listing → bidding → checkout
- Built Flutter UI with smooth state updates
- Tech: Flutter, Firebase

Phonetic-Aware Sentiment Analysis Model | [Github](#) |

- Built dual-channel BiLSTM (word + phoneme) for code-mixed text
- Designed phonetic pipeline (CMU G2P + ITRANS) to reduce OOV errors and added cross-modal attention for feature fusion
- Trained on 15K+ tweets; F1: 0.587, ROC-AUC: 0.761, Acc: 0.582, Ran ablation + noise tests; stable to F1 0.45 @ 50% noise
- Tech: Python, PyTorch, NLP

EDUCATION

Institute of Engineering and Management(IEM) |
Bachelor of Technology in Information Technology (CGPA: 8+)

Kolkata, India

July 2023- May 2027

B.D. Memorial International |
Class XII (CBSE) – 80%

Kolkata, India

April 2021-May 2023